

NORTH SYDNEY BOYS HIGH SCHOOL

2015 HSC ASSESSMENT TASK 2

Mathematics Extension 1

General Instructions

- Working time – 55 minutes (+ 5 minutes reading time)
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Ms Ziazaris
- ☐ Mr Zuber
- ☐ Mr Ireland
- ☐ Mr Lam
- ☐ Mr Lin

Student Number: _____

(To be used by the exam markers only.)

Question No	1-5	6	7	8	9	10	11	Total	Total %
Mark	$\overline{5}$	$\overline{7}$	$\overline{10}$	$\overline{4}$	$\overline{8}$	$\overline{7}$	$\overline{6}$	$\overline{47}$	$\overline{100}$

MULTIPLE CHOICE

Select the alternative A, B, C or D that best answers the question and indicate your choice by shading the appropriate letter.

1.	☐ A	☐ B	☐ C	☐ D
2.	☐ A	☐ B	☐ C	☐ D
3.	☐ A	☐ B	☐ C	☐ D
4.	☐ A	☐ B	☐ C	☐ D
5.	☐ A	☐ B	☐ C	☐ D

Objective Response

Mark your answers on the multiple choice box on the opposite page.

Marks

1 $\frac{\log_{12} 8}{\log_{12} 2}$ is equal to: 1

- (A) $\log_{12} 4$ (B) $\log_{12} 6$ (C) 3 (D) 0.5

2 The derivative with respect to x of $\log \sqrt{x^2 - 1}$ is: 1

- (A) $\frac{x}{\sqrt{x^2 - 1}}$ (B) $\frac{x}{x^2 - 1}$ (C) $\frac{x}{2(x^2 - 1)}$ (D) $\frac{x}{2\sqrt{x^2 - 1}}$

3 The area under the curve $y = \frac{5}{\sqrt{x}}$, for $1 \leq x \leq e^3$, is rotated about the x -axis.

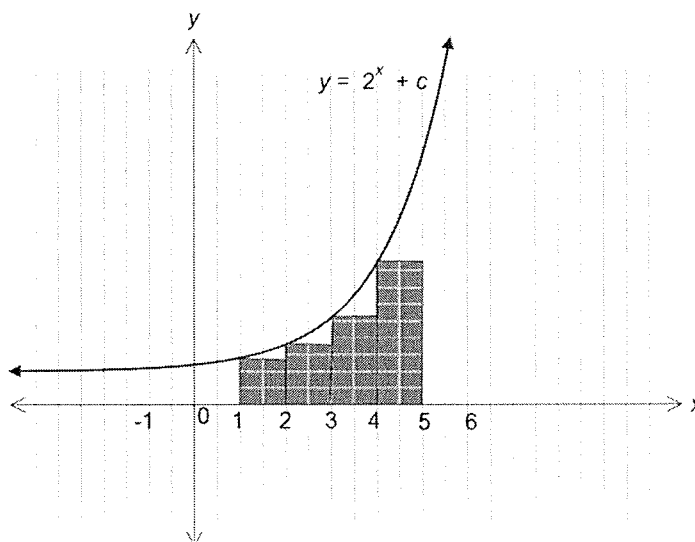
The volume of the solid generated is: 1

- (A) 15π cubic units (B) 25π cubic units
(C) 28π cubic units (D) 75π cubic units

4 If $\int_0^1 \frac{e^x}{1 + e^x} dx = \log_e K$, then K equals : 1

- (A) $1 + e$ (B) e (C) $\frac{e + 1}{2}$ (D) $\frac{(e + 1)^2}{2}$

5 Consider the graph $y = 2^x + c$, where c is a real number. The area of the shaded rectangles is used to find an **approximation** to the area of the region that is bounded by the graph, the x -axis and the lines $x = 1$ and $x = 5$.



If the total area of the shaded rectangles is 44, then the value of c is:

- (A) 14 (B) -4 (C) $\frac{14}{5}$ (D) $\frac{7}{2}$

1

Extended Response

Write your solutions in your exam booklet, starting a new page for each Question.

Marks

Question 6 (7 marks)

- (a) Find the coordinates of the point on the curve $y = 2e^{3x} + 1$ where the tangent is parallel to the line $12x - y + 1 = 0$. 3
- (b) Solve the equation $\log_3(2x - 1) + \log_3(x - 4) = 2$ 3
- (c) Write down the domain of the function $f(x) = \sqrt{-\log_e x}$ 1

Question 7 (10 marks)

- (a) Differentiate each of the following with respect to x :

- (i) $x^3 e^{-4x}$ 2
- (ii) $\sqrt{e^x}$ 1
- (iii) $\log_e \left(\frac{3+2x}{5-x} \right)$ 2

- (b) Find:

- (i) $\int \frac{3x}{4+x^2} dx$ 2
- (ii) $\int \sqrt{e^x} dx$ 1
- (iii) $\int x e^{x^2+2} dx$ 2

Question 8 (4 marks)

(i) Prove using mathematical induction that

$$(1 \times 2) + (2 \times 3) + (3 \times 4) + \dots + n(n+1) = \frac{1}{3}n(n+1)(n+2), \text{ for integers } n \geq 1. \quad 3$$

(ii) Hence evaluate $\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n k(k+1)$ 1

Question 9 (8 marks)

Given that $f(x) = \frac{e^x}{e^x + 1}$,

(i) Find the first derivative of $f(x)$, and hence show that $y = f(x)$ is a monotonic increasing function for all x . 2

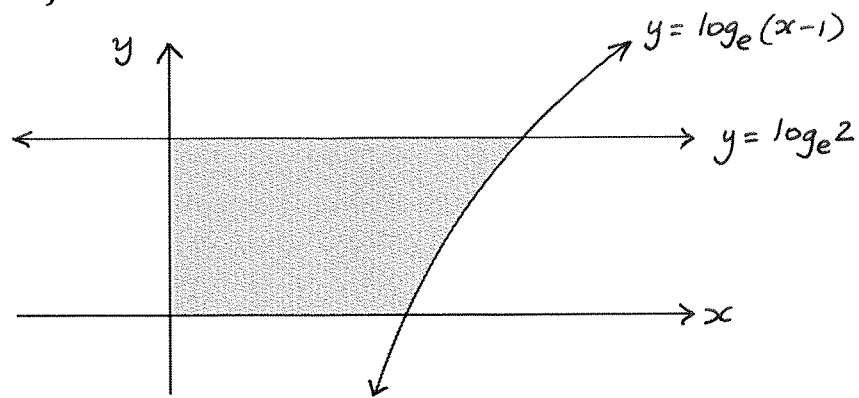
(ii) What is the behaviour of $f(x)$ as $x \rightarrow \infty$ and $x \rightarrow -\infty$? 2

(iii) Show that the second derivative of the function is $\frac{e^x(1-e^x)}{(e^x+1)^3}$ and hence find the coordinates of any points of inflexion. 2

(iv) Sketch the graph of $y = f(x)$, showing any intercepts, turning points, inflexions, or asymptotes. 2

Question 10 (7 marks)

(a)



Find the area enclosed by the curve $y = \log_e(x-1)$, the coordinate axes, and the line $y = \log_e 2$ (see diagram). 3

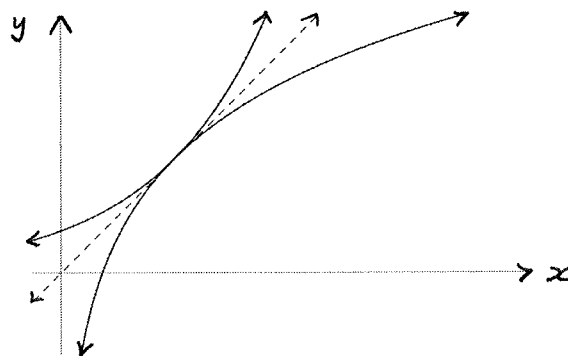
(b)

(i) Differentiate $y = x^2 \log x$ 1

(ii) Hence evaluate $\int_1^e x \log x \, dx$ 3

Question 11 (6 marks)

Two functions, $f(x) = a^x$ and $g(x) = \log_a x$, are drawn on the same axes so that they touch on the line $y = x$ (see diagram). Note that $a > 0$.



(i) Show that at the point where they touch $a^x \times \ln a = \ln x$. 1

(ii) Write expressions for $f'(x)$ and $g'(x)$. 2

(iii) Hence find the coordinates of the point of contact of $y = f(x)$ and $y = g(x)$. 2

(iv) What is the value of a ? 1

END OF EXAM

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

$$[1.] \quad \frac{\log 8}{\log 2} = \frac{3 \log 2}{\log 2} = 3 \quad \therefore (C) \quad \checkmark$$

$$[2.] \quad \frac{d}{dx} \log (x^2-1)^{\frac{1}{2}} = \frac{1}{\sqrt{x^2-1}} \cdot \frac{1}{2} \cdot 2x \cdot \frac{1}{\sqrt{x^2-1}} \\ = \frac{x}{x^2-1} \quad \therefore (B) \quad \checkmark$$

$$[3.] \quad V = \pi \int_1^{e^3} \frac{25}{x} dx \\ = 25\pi [\ln x]_1^{e^3} = 75\pi \quad \therefore (D) \quad \checkmark$$

$$[4.] \quad \int_0^1 \frac{e^x}{1+e^x} dx = \left[\ln(1+e^x) \right]_0^1 \\ = \ln(1+e) - \ln(1+1) \\ = \ln\left(\frac{1+e}{2}\right) \quad \therefore (C) \quad \checkmark$$

$$[5.] \quad \text{Area rectangles} = (2^1+c) + (2^2+c) + (2^3+c) \\ + (2^4+c) \\ = 2+4+8+16 + 4c \\ = 30 + 4c \\ \therefore 4c = 14, \quad \therefore c = \frac{7}{2} \quad \therefore (D) \quad \checkmark$$

Q6

$$(a) \quad y = 2e^{3x} + 1$$

$$\therefore y' = 6e^{3x}$$

$$\text{Also, as } y = 12x + 1 \quad \therefore m = 12$$

$$\therefore 6e^{3x} = 12$$

$$\therefore e^{3x} = 2 \quad \therefore x = \frac{\ln 2}{3}$$

$$\text{So } \therefore y = 2 \cdot e^{3 \cdot \frac{\ln 2}{3}} + 1 = 2 \cdot 2 + 1 = 5$$

$$\therefore \left(\frac{\ln 2}{3}, 5 \right)$$

✓ For this equality

✓✓ one each coordinate

$$(b) \quad \log_3(2x-1) + \log_3(x-4) = 2$$

$$\therefore \log_3[(2x-1)(x-4)] = 2$$

$$(2x-1)(x-4) = 9$$

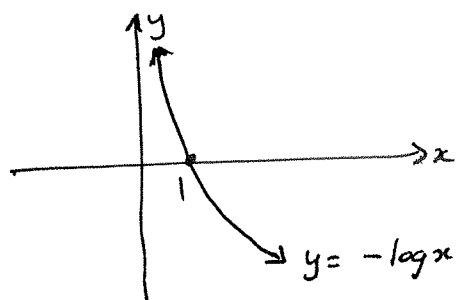
$$2x^2 - 9x - 5 = 0$$

$$(2x+1)(x-5) = 0$$

$$\therefore x = -\frac{1}{2} \text{ or } x = 5$$

$$\text{But } \log_3\left(-\frac{1}{2}-4\right) \text{ not defined, } \therefore x = 5.$$

$$(c) \quad f(x) = \sqrt{-\log x}$$



$$\therefore D: 0 < x \leq 1.$$

✓

7

Q7

(a) (i) $y = x^3 e^{-4x}$

$$\therefore y' = e^{-4x} \cdot 3x^2 + x^3 \cdot -4e^{-4x}$$

$$= x^2 e^{-4x} (3 - 4x)$$

(ii) $y = \sqrt{e^x} = (e^x)^{\frac{1}{2}} = e^{\frac{x}{2}}$

$$\therefore y' = \frac{1}{2} e^{\frac{x}{2}}$$

$$= \frac{1}{2} \sqrt{e^x}$$

(iii) $y = \ln\left(\frac{3+2x}{5-x}\right)$

$$= \ln(3+2x) - \ln(5-x)$$

$$\therefore y' = \frac{2}{3+2x} + \frac{1}{5-x}$$

(b) (i) $\int \frac{3x}{4+x^2} dx = \frac{3}{2} \int \frac{2x}{4+x^2} dx$

$$= \frac{3}{2} \ln(4+x^2) + C$$

(ii) $\int \sqrt{e^x} dx = \int e^{\frac{x}{2}} dx$

$$= 2e^{\frac{x}{2}} + C$$

$$= 2\sqrt{e^x} + C$$

(iii) $\int x e^{x^2+2} dx = \frac{1}{2} e^{x^2+2} + C$

✓✓

✓

✓

✓

✓✓

✓

✓✓

[delete 1
if no
constant]

Q8 (i) When $n=1$:

$$\text{LHS} = 1(1+1) = 2$$

$$\text{RHS} = \frac{1}{3}(1)(1+1)(\cancel{1+2}) = 2 = \text{LHS}$$

$\therefore$ true for $n=1$

✓ must
sub in

Assume true for $n=k$:

i.e. assume $1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) = \frac{1}{3} \cdot k(k+1)(k+2)$

Then we'd have

$$\begin{aligned} 1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) + (k+1)(k+2) \\ = \frac{1}{3} \cdot k(k+1)(k+2) + (k+1)(k+2) \end{aligned}$$

by assumption

✓

$$= (k+1)(k+2) \left[\frac{1}{3} \cdot k + 1 \right]$$

$$= (k+1)(k+2) \cdot \frac{1}{3} [k+3]$$

$$= \frac{1}{3} \cdot (k+1)(k+2)(k+3)$$

✓

$\therefore$ if true for $n=k$, it's true for $n=k+1$

$\therefore$ Since true for $n=1$ it is true for all $n \geq 1$ by the principle of mathematical induction.

(ii) $\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n k(k+1) = \lim_{n \rightarrow \infty} \frac{\frac{1}{3} n(n+1)(n+2)}{n^3}$ from (i)

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{3} \cdot \frac{n}{n} \cdot \frac{n+1}{n} \cdot \frac{n+2}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{3} \cdot 1 \cdot \left(1 + \frac{1}{n}\right) \cdot \left(1 + \frac{2}{n}\right) \right]$$

$$= \frac{1}{3}$$

✓

Q9

$$f(x) = \frac{e^x}{e^x + 1}$$

$$(i) \therefore f'(x) = \frac{(e^x + 1) \cdot e^x - e^x \cdot e^x}{(e^x + 1)^2}$$

$$\therefore f'(x) = \frac{e^x}{(e^x + 1)^2}$$

Since $e^x > 0$ for all x , and $(e^x + 1)^2 > 0$ for all x ,

$\therefore f'(x) > 0$ for all $x \therefore f(x)$ is monotonic increasing.

(ii)

$$f(x) = \frac{e^x}{e^x + 1} = \frac{e^x + 1 - 1}{e^x + 1} = 1 - \frac{1}{e^x + 1}$$

$$\therefore \text{as } x \rightarrow \infty, e^x \rightarrow \infty \therefore f(x) \rightarrow 1^-$$

$$\text{as } x \rightarrow -\infty, e^x \rightarrow 0 \therefore f(x) \rightarrow 0^+$$

$$(iii) f'(x) = \frac{e^x}{(e^x + 1)^2}$$

$$\therefore f''(x) = \frac{(e^x + 1)^2 \cdot e^x - e^x \cdot 2 \cdot e^x (e^x + 1)}{(e^x + 1)^4}$$

$$= \frac{(e^x + 1) \cdot e^x - 2e^{2x}}{(e^x + 1)^3}$$

$$= \frac{e^x - e^{2x}}{(e^x + 1)^3}$$

$$\therefore f''(x) = \frac{e^x (1 - e^x)}{(e^x + 1)^3}$$

✓ (no fudging!)

→ P.T.D.

Q9 - continued

(iii) - continued:

For inflexions, $f''(x) = 0$

$$\therefore e^x = 1 \quad \therefore \quad \begin{matrix} x = 0 \\ y = \frac{1}{2} \end{matrix}$$

Test:

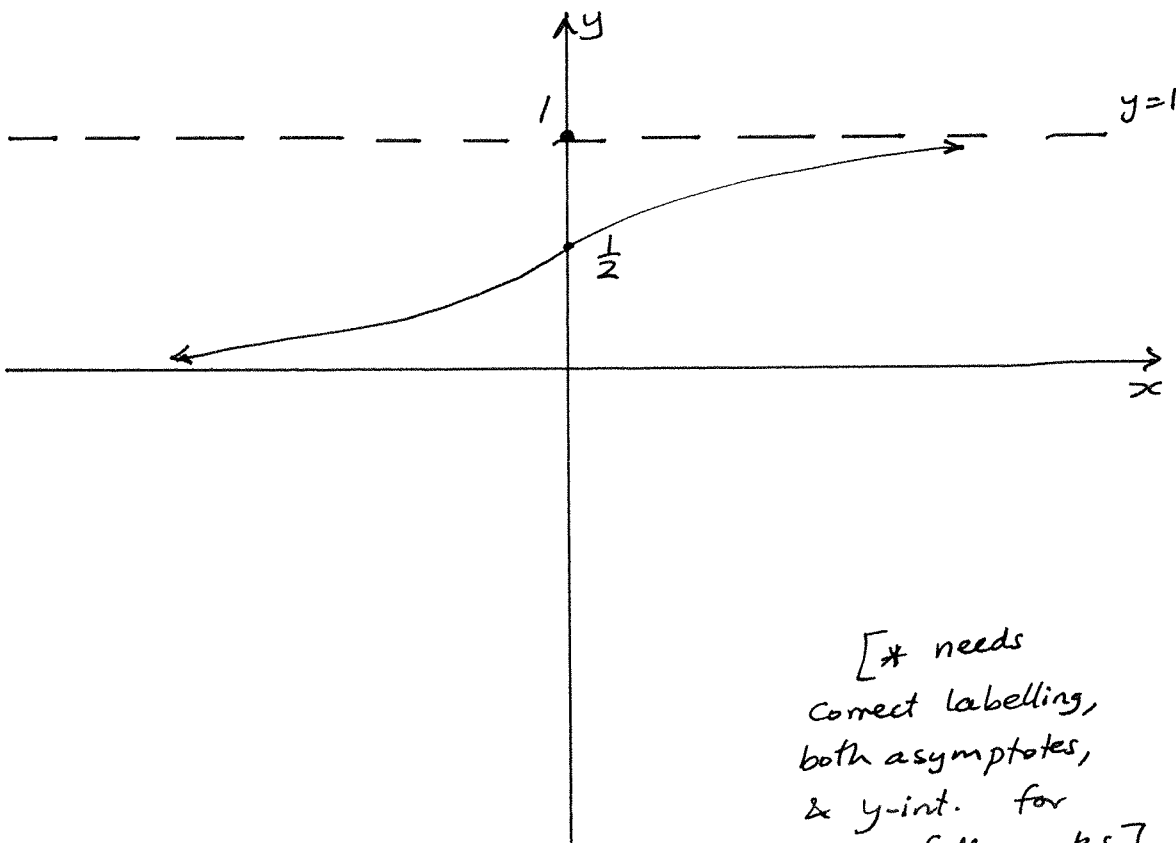
x	$-\ln 2$	0	$\ln 2$
$f''(x)$	$\frac{(\frac{1}{2})(1-\frac{1}{2})}{(\frac{1}{2}+1)^3}$	0	$\frac{2(1-2)}{(2+1)^3}$
	$+$	0	$-$

Change of concavity

 $\therefore (0, \frac{1}{2})$ is inflexion.needs both
coords &
some kind
of test.

✓

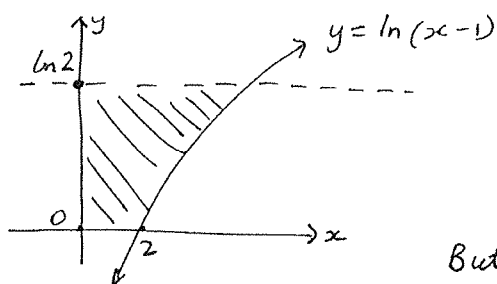
(iv)

[* needs
correct labelling,
both asymptotes,
& y-int. for
full marks]

✓✓

Q10

(a)



$$A = \int_0^{\ln 2} x \, dy$$

But $y = \log_e(x-1)$

$$\therefore x = e^y + 1$$

$$\therefore A = \int_0^{\ln 2} (e^y + 1) \, dy$$

$$= [e^y + y]_0^{\ln 2}$$

$$= (e^{\ln 2} + \ln 2) - (e^0 + 0)$$

$$\therefore A = 1 + \ln 2 \text{ units}^2.$$

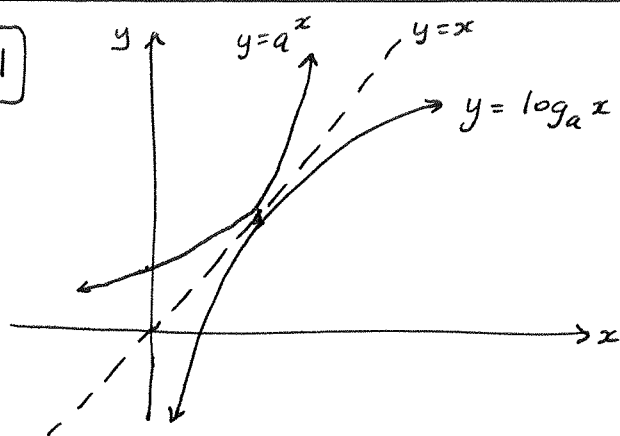
(b) (i) $\left. \begin{aligned} \frac{d}{dx} (x^2 \log x) &= \log x \cdot 2x + x^2 \cdot \frac{1}{x} \\ &= 2x \log x + x \end{aligned} \right\}$

(ii) From (i),

$$\int \frac{d}{dx} (x^2 \log x) \, dx = \int 2x \log x \, dx + \int x \, dx$$

$$\therefore \int_1^e x \log x \, dx = \frac{1}{2} \left[\int_1^e \frac{d}{dx} (x^2 \log x) \, dx - \int_1^e x \, dx \right] \left. \begin{aligned} &= \frac{1}{2} \left[[x^2 \log x]_1^e - \left[\frac{x^2}{2} \right]_1^e \right] \\ &= \frac{1}{2} \left[(e^2 \cdot \log e - 1^2 \cdot \log 1) - \left(\frac{e^2}{2} - \frac{1}{2} \right) \right] \\ &= \frac{1}{2} \left(e^2 - \frac{e^2}{2} + \frac{1}{2} \right) \\ &= \frac{e^2 + 1}{4} \quad (\div 2.097 \, 264) \end{aligned} \right\}$$

Q11



(i) when they touch, $a^x = \log_a x$
 $= \frac{\ln x}{\ln a}$

$$\therefore a^x \cdot \ln a = \ln x.$$

(ii) $f(x) = a^x = (e^{\ln a})^x \therefore f'(x) = \ln a \cdot a^x$

$$g(x) = \log_a x = \frac{\ln x}{\ln a} \therefore g'(x) = \frac{1}{\ln a \cdot x}$$

(iii) They touch on $y = x$, $\therefore f' = g' = 1$

Since $f'(x) = 1 \therefore \ln a \cdot a^x = 1 \therefore a^x = \frac{1}{\ln a}$

But $a^x = \frac{\ln x}{\ln a}$ (from i))

$$\therefore \frac{1}{\ln a} = \frac{\ln x}{\ln a} \therefore \ln x = 1$$

$$x = e$$

$$y = e$$

So point contact is (e, e) .

(iv) Since $g'(e) = 1 \therefore \frac{1}{\ln a \cdot e} = 1$

$$\therefore \ln a = \frac{1}{e} \therefore a = e^{\frac{1}{e}}$$

(other routes/working possible).